The empirical recurrence for OEIS A239821: recovering a conjecture that is not written down

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Abstract

OEIS A239821 records "Empirical recurrence of order 64", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 64$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 64, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 64 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A239821 is "Number of 3Xn 0..3 arrays with no element equal to zero plus the sum of elements to its left or one plus the sum of the elements above it, modulo 4.". It has offset 1 and begins

$$11, 113, 1480, 18728, 232272, 2912793, 36627126, 459870638, \ \ldots$$

The entry states:

> Empirical recurrence of order 64 (see link above)

As of the "Last modified" line on the live entry (Jun 02 2025 09:32:42, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 64$ vertices, $a(n) = \iota^\top M^{\,j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 64$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 128$ exact terms of the model returns order 64 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{64} c_i\, a(n-i),$$

c_{1}	27	c_{17}	3341168768	c_{33}	1036894753634	c_{49}	858815115360
c_{2}	−302	c_{18}	−4856558327	c_{34}	67292173294	c_{50}	−243440086688
c_{3}	2422	c_{19}	7687046664	c_{35}	−1481629252600	c_{51}	−573204975680
c_{4}	−16437	c_{20}	15377593503	c_{36}	−9389133976304	c_{52}	1060174640768
c_{5}	85302	c_{21}	−19064113799	c_{37}	1456577954039	c_{53}	44140228096
c_{6}	−358454	c_{22}	−5822039996	c_{38}	4726042505038	c_{54}	−723018236416
c_{7}	1274249	c_{23}	−87114885857	c_{39}	3717186665863	c_{55}	125018898432
c_{8}	−3235456	c_{24}	9475583786	c_{40}	9492546947253	c_{56}	22933151744
c_{9}	5269788	c_{25}	99983482224	c_{41}	−7811914846060	c_{57}	9658417152
c_{10}	−1518156	c_{26}	253408323168	c_{42}	−8991320234906	c_{58}	142179696640
c_{11}	−38839230	c_{27}	357071016619	c_{43}	4149163287848	c_{59}	−6200295424
c_{12}	146076164	c_{28}	−706386429046	c_{44}	−1762717295764	c_{60}	−42433773568
c_{13}	−268326361	c_{29}	−455088546671	c_{45}	538131665000	c_{61}	−1030750208
c_{14}	427237294	c_{30}	−925106026875	c_{46}	4268454871944	c_{62}	2172649472
c_{15}	60714039	c_{31}	−465440690648	c_{47}	−449020960568	c_{63}	268435456
c_{16}	−2362109717	c_{32}	3977062456415	c_{48}	−2360874285568	c_{64}	−67108864

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 64, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $64 + 4$ terms removed returns order 64 as well, so $\deg q_{\min} = 64 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $64 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 64 that this sequence satisfies — and that family turns out to have exactly one member.

References

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